

# LO TRÌNH 6 THANG TRỞ THÀNH

## CLOUD ENGINEER

Phân tích 7 Roadmap PDFs (roadmap.sh) + Thị trường Việt Nam 2025-2026

| THÔNG TIN   | CHI TIẾT                                                         |
|-------------|------------------------------------------------------------------|
| Thời gian   | 6 tháng (26 tuần)                                                |
| Phương pháp | Hybrid: Job-task-first + Project-first + Lab thực hành           |
| Mục tiêu    | Junior Cloud/DevOps Engineer HCM — 15-25M VND                    |
| Chứng chỉ   | AWS Solutions Architect Associate SAA-C03                        |
| Portfolio   | 3 GitHub repos + Runbooks + Architecture Diagrams                |
| Tools đã có | WSL2/Ubuntu, Docker, AWS CLI 2.34, Terraform 1.15, kubectl, Helm |

### 1. YÊU CẦU THỊ TRƯỜNG VIỆT NAM 2025-2026

Phân tích JD từ ITviec, TopDev, LinkedIn — TP.Ho Chi Minh:

| Kỹ năng / Công nghệ                 | Tần suất JD | Mức độ     |
|-------------------------------------|-------------|------------|
| Linux (Ubuntu/CentOS/RHEL)          | 90%         | Bắt buộc   |
| Git / GitHub                        | 95%         | Bắt buộc   |
| Docker & Containers                 | 85%         | Bắt buộc   |
| AWS (EC2, VPC, IAM, S3, CloudWatch) | 80%         | Bắt buộc   |
| CI/CD (GitHub Actions / Jenkins)    | 75%         | Bắt buộc   |
| Networking (TCP/IP, DNS, CIDR, SSH) | 70%         | Quan trọng |
| Kubernetes (EKS/k8s)                | 65%         | Quan trọng |
| Terraform / IaC                     | 60%         | Quan trọng |
| Python scripting                    | 55%         | Quan trọng |
| Monitoring (Prometheus + Grafana)   | 50%         | Nên có     |
| Ansible / Config Management         | 40%         | Thêm điểm  |

| Level               | Kinh nghiệm                     | Luong HCM    |
|---------------------|---------------------------------|--------------|
| Junior Cloud/DevOps | 0-1 năm + SAA cert + 3 projects | 15 - 22M VND |
| Mid-level           | 1-3 năm                         | 25 - 40M VND |
| Senior              | 3+ năm                          | 40 - 70M VND |

### 2. THU TU HỌC TỐI ƯU TỪ 7 ROADMAP PDFs

| Bước | Công nghệ                                                     | Nguồn PDF               | Tuần     |
|------|---------------------------------------------------------------|-------------------------|----------|
| 1    | Linux Fundamentals (Shell, Process, Service, Networking)      | linux.pdf               | Week 1-2 |
| 2    | Networking & Protocols (TCP/IP, DNS, SSH, HTTP/S, Subnetting) | networking-engineer.pdf | Week 2   |
| 3    | Git, Bash Scripting, Python cơ bản                            | devops.pdf              | Week 3   |

| Buoc | Cong nghe                                                      | Nguon PDF                | Tuan       |
|------|----------------------------------------------------------------|--------------------------|------------|
| 4    | Docker (Dockerfile, Compose, Registry, Security, multi-stage)  | docker.pdf               | Week 4     |
| 5    | AWS Core Services (IAM -> VPC -> EC2 -> S3 -> CloudWatch)      | aws.pdf                  | Week 5-8   |
| 6    | CI/CD — GitHub Actions                                         | devops.pdf               | Week 10    |
| 7    | Terraform — Infrastructure as Code                             | devops.pdf               | Week 9-12  |
| 8    | Kubernetes (K8s/EKS + Helm + GitOps + Autoscaling)             | kubernetes.pdf           | Week 13-16 |
| 9    | Monitoring (Prometheus, Grafana, Loki, CloudWatch)             | devops.pdf               | Week 17    |
| 10   | Security (IAM least-privilege, RBAC, Secrets, Container Scans) | aws.pdf + kubernetes.pdf | Week 18    |
| 11   | AWS Certification SAA-C03 (review all above)                   | all                      | Week 21-23 |

### 3. LO TRÌNH CHI TIẾT 6 THANG

#### THANG 1: NEN TANG HE THONG (Weeks 1-4)

Mục tiêu: Vung Linux, Networking, Git, Docker — nen tang bat buoc

| Tuan   | Chu de                                                                                | Lab thuc hanh                                                                                                       | Output                                              |
|--------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| Week 1 | Linux Fundamentals<br>(Navigation, Permissions, Process, Service, Package Mgmt)       | 50+ lenh WSL Ubuntu<br>Tao systemd service<br>Cai + cau hinh Nginx                                                  | linux-admin-labs/week1/<br>runbook.md               |
| Week 2 | Linux Advanced + Networking<br>(Disk/LVM, Shell Script, SSH, DNS, TCP/IP, Subnetting) | Script tu dong backup<br>SSH key-based auth<br>Debug website down                                                   | linux-admin-labs/week2/<br>scripts/ + runbooks/     |
| Week 3 | Git + Bash + Python co ban<br>(branch, merge, conflict, scripting, cron, boto3)       | Tao repo cloud-job-task-labs<br>Script health-check cron<br>Python goi API + JSON                                   | cloud-job-task-labs/ repo<br>README.md              |
| Week 4 | Docker Fundamentals<br>(Dockerfile, Compose, Registry, Security)                      | Docker compose: FastAPI + PostgreSQL<br>Build multi-stage image<br>multi-stage build<br>Trivy scan + push DockerHub | task-04-docker-deploy/<br>dockerfile + compose.yaml |

#### THANG 2: AWS CORE SERVICES (Weeks 5-8)

Mục tiêu: Thanh thao IAM, VPC, EC2, S3, CloudWatch, Route53, RDS, ECS

| Tuan   | Chu de                                                                                 | Lab                                                                                       | Output                                              |
|--------|----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|-----------------------------------------------------|
| Week 5 | AWS Setup + IAM + VPC<br>(MFA, billing alarm, Users/Roles, CIDR, Subnets, IGW, NAT GW) | Setup account an toan<br>Tao VPC 10.0.0.0/16<br>Diagram architecture                      | VPC diagram draw.io<br>IAM policy review            |
| Week 6 | EC2 + S3 + Deploy<br>(Instance types, Elastic IP, AMI, ASG, S3 storage classes)        | Launch EC2, SSH vao<br>Deploy Nginx + FastAPI<br>Boto3 backup -> S3                       | task-10-aws-ec2-deploy/<br>Backup automation script |
| Week 7 | Route53 + CloudWatch + CloudFront<br>(DNS, Metrics, Logs, Alarms, CDN, HTTPS/ACM)      | Setup domain + HTTPS<br>CloudWatch alarm CPU>80%<br>Debug 502/504 scenario                | task-07-nginx-502-debug/<br>Post-mortem report      |
| Week 8 | RDS + ECS + PROJECT 1<br>(RDS PostgreSQL, ECS Fargate, ECR, GitHub Actions pipeline)   | ECS Fargate deploy<br>RDS private subnet<br>GitHub Actions pipeline<br>PROJECT 1 complete | aws-docker-webapp-deployment<br>(repo hoan chinh)   |

| AWS Service     | Chi phi uoc tinh | Ghi chu                             |
|-----------------|------------------|-------------------------------------|
| EC2 t2.micro    | \$0              | Free tier 12 thang dau              |
| RDS db.t3.micro | \$0              | Free tier 750h/thang                |
| S3 (5GB)        | < \$1            | Gan nhu mien phi                    |
| ECS Fargate     | ~\$5-15/thang    | Chi chay khi lab, destroy xong      |
| EKS Cluster     | ~\$7/ngay        | Tao -> dung -> destroy ngay sau lab |
| TONG uoc tinh   | ~\$20-40/thang   | Neu destroy sau moi lab             |

QUAN TRONG: Luon terraform destroy sau lab. Dat billing alarm \$20. Khong dung root account.

#### THANG 3: TERRAFORM + CI/CD + ANSIBLE (Weeks 9-12)

| Tuan    | Chu de                                                                                       | Lab                                                                                 | Output                                         |
|---------|----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------|
| Week 9  | Terraform Co ban<br>(providers, state, modules, S3 backend, Dynamic Block), terraform import | S3 bucket -> EC2+VPC+SG -> ECS<br>DynamoDB, terraform import                        | modules/vpc + modules/ec2                      |
| Week 10 | GitHub Actions CI/CD<br>(secrets, multi-env, Terraform pipeline)                             | CI: build+test Docker<br>CD: push ECR + deploy ECS<br>Infra: PR->plan, merge->apply | github-actions-cloud-cicd/ repo                |
| Week 11 | Nginx + Ansible<br>(reverse proxy, HTTPS Certbot, playbooks, debug flags)                    | Nginx HTTPS + Ansible deploy<br>Setup Nginx loi pho bien                            | task-06-nginx/<br>ansible-playbooks/           |
| Week 12 | PROJECT 2: Terraform AWS Infra<br>(full modules + CI/CD infra pipeline)                      | VPC+EC2+RDS complete<br>GitHub Actions plan->apply<br>README + architecture diagram | terraform-aws-cloud-infra<br>(repo hoan chinh) |

## THANG 4: KUBERNETES — K8s + EKS (Weeks 13-16)

| Tuan    | Chu de                                                                                | Lab                                                                                                | Output                                                |
|---------|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| Week 13 | K8s Fundamentals<br>(Pods, Deployments, Services, ConfigMaps, Secrets, kubectl)       | kind/minikube local cluster<br>Deploy 3-replica nginx<br>ConfigMap + Secret inject                 | k8s-labs/ YAML files                                  |
| Week 14 | K8s Storage + Security + Monitoring<br>(PVC, StatefulSet, RBAC, Network Policy)       | PostgreSQL StatefulSet<br>RBAC service account<br>HPA, Prometheus<br>Helm install prometheus-stack | monitoring/ namespace setup                           |
| Week 15 | Helm + GitOps + Deploy Patterns<br>(Helm charts, ArgoCD, Rolling, Canary, Blue-Green) | Custom Helm chart FastAPI<br>ArgoCD sync Git -> K8s<br>Canary 10%->50%->100%                       | helm-charts/ + argocd-apps/                           |
| Week 16 | EKS + PROJECT 3<br>(eksctl, IRSA, ALB Ingress, Cluster Autoscaler)                    | eksctl create cluster<br>Deploy production app<br>Full observability stack                         | kubernetes-eks-production-deploy<br>(repo hoan chinh) |

## THANG 5: MONITORING + SECURITY + SRE (Weeks 17-20)

| Tuan    | Chu de                                                                                   | Lab                                                                                      | Output                                          |
|---------|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|-------------------------------------------------|
| Week 17 | Monitoring & Observability<br>(Prometheus, Grafana, Loki, Alertmanager, OpenTelemetry)   | Dashboard EC2 + K8s metrics<br>Ship logs Loki -> Grafana<br>Alert Slack khi service down | monitoring-stack/ setup<br>SLI/SLO definition   |
| Week 18 | Security Hardening<br>(IAM least privilege, GuardDuty, Secrets Manager, Trivy, K8s RBAC) | Migrate secrets -> SSM<br>Scan tat ca images<br>Apply Pod Security Admission             | Security checklist<br>Hardened infra configs    |
| Week 19 | Advanced CI/CD + GitOps<br>(Multi-env pipeline, Sealed Secrets, FluxCD, Jenkins)         | dev/staging/prod pipeline<br>Sealed Secrets trong Git<br>Jenkinsfile + GitLab CI         | Multi-env pipeline<br>.gitlab-ci.yml example    |
| Week 20 | SRE Practices + DR<br>(SLI/SLO, Incident runbooks, Chaos Engineering, Backup/Restore)    | Define SLO cho apps<br>Kill pod check recovery<br>RDS shapshot restore test              | SRE runbook template<br>DR architecture diagram |

## THANG 6: AWS CERTIFICATION + JOB PREP (Weeks 21-26)

| Tuan       | Chu de                                                                       | Tai lieu / Lab                                                                     | Muc tieu                                        |
|------------|------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------|
| Week 21-23 | AWS SAA-C03<br>(EC2+Storage, VPC+HA, Serverless, Security, Well-Architected) | Stephane Maarek (Udemy)<br>Tutorials Dojo Practice Exams<br>AWS Skill Builder labs | Practice exam >= 82%<br>Dang ky thi Pearson VUE |
| Week 24    | Portfolio Polishing<br>(README, diagrams, blogs, demo videos)                | draw.io + AWS Architecture Icons<br>yiblo.asia / dev.to blogs                      | 3 repos hoan chinh<br>2+ bai viet ky thuat      |
| Week 25    | CV + LinkedIn + Apply<br>(1-trang CV, ITviec, TopDev, LinkedIn OpenApply)    | CV technical template<br>Apply 5 JD/ngay                                           | Apply >= 30 vi tri<br>Target: 18-22M VND        |
| Week 26    | Phong van ky thuat<br>(50 cau technical + behavioral, system design co ban)  | Mock interview<br>Whiteboard VPC design<br>Debug scenarios                         | Job offer 18-22M VND                            |

| Tuan    | Chu de SAA-C03                                                             | Practice Target         |
|---------|----------------------------------------------------------------------------|-------------------------|
| Week 21 | EC2 advanced, EBS/S3/EFS/Glacier, RDS/Aurora/DynamoDB/ElastiCache          | Quiz theo tung service  |
| Week 22 | ALB/NLB/ASG HA, VPC/Direct Connect/VPN, CloudFront/Route53, Lambda/SQS/SNS | Mock exam 30 cau        |
| Week 23 | IAM/KMS/SSM/Shield/WAF, Cost Optimization, Well-Architected Framework      | Full exam 65 cau >= 82% |

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## 4. PORTFOLIO PROJECTS — 3 GITHUB REPOS

| Project                              | Tech Stack                                                                                        | Highlights                                                                     | Thang                |
|--------------------------------------|---------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|----------------------|
| aws-docker-webapp-deployment         | FastAPI + PostgreSQL + Docker<br>ECS Fargate + ALB + ACM<br>RDS + CloudWatch + GitHub Actions     | Full HTTPS, auto CI/CD<br>CloudWatch monitoring<br>Architecture diagram day du | Thang 2<br>(Week 8)  |
| terraform-aws-cloud-infra            | Terraform modules (VPC/EC2/RDS/S3)<br>S3 backend + DynamoDB lock<br>GitHub Actions infra pipeline | Infra hoàn toàn idempotent<br>Modules có document<br>Infra PR review workflow  | Thang 3<br>(Week 12) |
| kubernetes-eks-production-deployment | EKS + Helm + ArgoCD GitOps<br>Prometheus + Grafana + Loki<br>HPA + Cluster Autoscaler             | GitOps workflow<br>Auto-scaling full stack<br>Production observability         | Thang 4<br>(Week 16) |

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## 5. CAU HOI PHONG VAN KY THUAT (50 CAU)

### Linux

Khi server load cao, debug the nao? (uptime, top, sar, vmstat, iostat)  
Hard link va soft link khác nhau the nao?  
systemd vs SysV init — when to use which?  
Inode là gì? Tại sao "no space left" dù df còn trong?  
Xem process đang dùng port nào? (ss -tulnp, lsof -i)

### Docker

COPY vs ADD trong Dockerfile khác nhau the nao?  
Docker network modes: bridge vs host vs overlay vs macvlan?  
Làm sao giảm image size? (multi-stage, .dockerignore, slim base image)  
Docker compose vs Docker Swarm — dùng cái nào khi nào?  
Container bị OOMKilled — debug the nao?

### AWS

Giải thích VPC architecture của Project 1 của bạn?  
NAT Gateway vs NAT Instance — khác nhau cho nào?  
ALB vs NLB — khi nào dùng cái nào?  
S3 storage classes (Standard, IA, Glacier) — use case?  
IAM Role vs IAM User — best practice là gì?  
Làm sao thiết kế HA cho web app? (Multi-AZ, ASG, ALB)  
Security Group vs NACL — khác nhau the nao?  
Khi EC2 không thể ra internet, debug the nao?

### Terraform

terraform plan vs apply vs destroy?  
Xử lý state conflict the nao? (terraform state rm, mv)  
Module là gì, tại sao nên dùng?  
Remote state với S3 + DynamoDB lock hoạt động the nao?  
terraform import dùng khi nào?  
Terraform workspace vs separate state files?

### Kubernetes

Pod vs Deployment vs StatefulSet — khi nào dùng cái nào?  
Pod CrashLoopBackOff — debug the nao? (kubectl logs, describe, exec)  
HPA hoạt động the nao? Metrics source (Metrics Server)?

ClusterIP vs NodePort vs LoadBalancer vs Ingress?

RBAC: Role vs ClusterRole vs RoleBinding?

Helm chart là gì, tại sao dùng so với raw YAML?

ArgoCD/GitOps pattern hoạt động thế nào?

Pod không schedule được — debug thế nào? (node resources, taints)

## CI/CD & General

Giải thích pipeline CI/CD Project 2 từ đầu đến cuối?

Zero-downtime deployment — các cách tiếp cận?

Blue-green vs Canary deployment khác nhau thế nào?

Secret trong Git — làm sao xử lý an toàn? (Sealed Secrets, Vault)

SLI, SLO, SLA, Error Budget — ví dụ cụ thể?

Khi nào dùng ECS Fargate vs EKS?

Cost optimization trên AWS — bạn đã làm gì trong projects?

## Behavioral

Ke ve lan ban debug su co production nghiêm trong nhất?

Bạn học một công nghệ mới như thế nào? Ví dụ?

Tại sao bạn muốn chuyển sang Cloud/DevOps Engineering?

Khi team không đồng ý giải pháp của bạn, xử lý thế nào?

## 6. JOB-READY CHECKLIST

| Hang muc         | Tieu chi                                                        | Deadline   |
|------------------|-----------------------------------------------------------------|------------|
| Certification    | AWS SAA-C03 passed (score >= 750/1000)                          | Tuan 23    |
| GitHub Project 1 | aws-docker-webapp-deployment — README day du + CI/CD hoat dong  | Tuan 8     |
| GitHub Project 2 | terraform-aws-cloud-infra — modules documented + infra pipeline | Tuan 12    |
| GitHub Project 3 | kubernetes-eks-production-deploy — GitOps + monitoring stack    | Tuan 16    |
| Tech Writing     | 2+ bai blog tren Viblo.asia hoac dev.to                         | Tuan 24    |
| GitHub Profile   | README ca nhan + pinned repos + contribution graph              | Tuan 24    |
| CV               | 1 trang, English, skills + projects noi bat                     | Tuan 25    |
| LinkedIn         | Headline + Projects section + Open to Work                      | Tuan 25    |
| Job Applications | Apply >= 30 vi tri Cloud/DevOps Junior HCM                      | Tuan 25-26 |
| Mock Interviews  | Tra loi tu tin 50 cau trong phan phong van                      | Tuan 26    |

## 7. TAI LIEU HOC TAP

| Loai     | Tai lieu                                                 | Ghi chu                      |
|----------|----------------------------------------------------------|------------------------------|
| MIEN PHI | roadmap.sh (7 PDFs da co)                                | Da phan tich — su dung ngay  |
| MIEN PHI | AWS Free Tier + AWS Skill Builder                        | Official hands-on labs       |
| MIEN PHI | KodeKloud Play With Kubernetes                           | K8s playground mien phi      |
| MIEN PHI | GitHub Student Pack                                      | Free tools + credits         |
| MIEN PHI | YouTube: TechWorld with Nana, NetworkChuck, freeCodeCamp | Tot hiep YouTube DevOps      |
| MIEN PHI | Viblo.asia — cong dong IT Vietnam                        | Doc + viet bai ky thuat      |
| CO PHI   | Stephane Maarek AWS SAA (Udemy ~400K)                    | TÓT NHAT — sale thuong xuyen |
| CO PHI   | Tutorials Dojo Practice Exams SAA-C03 (~200K)            | Practice exam chuan nhat     |
| CO PHI   | KodeKloud DevOps Labs (~300K/thang)                      | Tot cho K8s lab co huong dan |

Lo trinh xay dung tu: 7 Roadmap PDFs (roadmap.sh) + Phan tich thi truong Vietnam 2025-2026. Cap nhat: 2026-08.